Overview

This document references the local control protocol using the device's TCP port. This protocol supports the following connection method:

- Local WiFi network communication using TCP/IP commands

This protocol is a two-way communication and control protocol for exclusive use with our Neo Smart Blinds Controller. All requests and commands are followed by an echo response indicating the command was successfully received. In this version, additional information concerning the reason for failing to execute a request is not returned.

Before using the local control communication, set up the controller and blinds using our "Neo Smart Blinds" app found at [Google Play]and [iOS App Store]. Follow the in-app instruction for adding the controller(s) and blind(s).

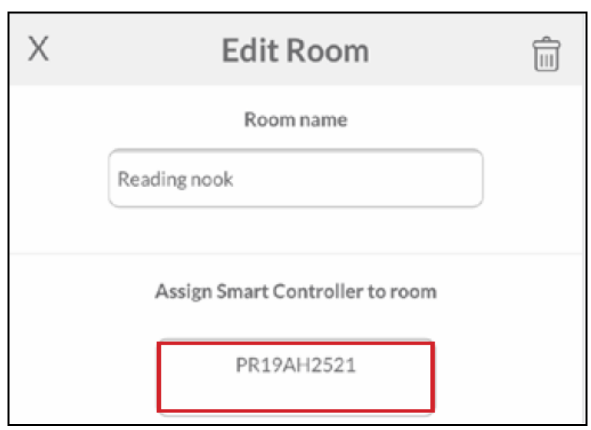
Blinds can be individually or group/room controlled (max 15 per group).

1. Protocol Conventions

Commands are case-insensitive. All commands and their responses are terminated with a carriage return/line feed pair (CRLF).

Local connection requires previous knowledge of IP and port addresses.

- Controller local IP can be found using the Neo Smart Blinds app, at "Your Controllers". For accounts with more than one controller make sure to use the IP address from the controller assigned to the room where the respective blind is located.
- Use port number 8839 (for TCP communication).
- The controller's channel and ids can be found on each blinds' advanced page using the Neo Smart Blinds app.



Each new request/command should open a new connection between the client (app or other devices) and the server (controller). The server will close the connection after every single request/command.

2. Script implementation

When implementing this protocol to an automated script some considerations need to be taken into account.

- The TCP server can only handle as many as 1-3 simultaneous client connections at most. However, it's highly recommended to only try one connection at a time.
- Each new client connection is closed by the server as soon as the message is received and replied to. That means before the command is validated and finally transmitted. That explains why only an echo message is returned to the client.
- Each received command takes at least 500ms to be verified, processed and transmitted by the controller/hub. If multiple requests are necessary, the script must implement a delay of at least 500ms between each TCP connection and command transmission.

3. TCP "transmit" function

variable type	: string
length	: 16 ASCII characters
message sent	: "**ID1.ID2-CHANNEL-ACTION!MOTORCODE" + CRLF**
returns	: if the message received: echo the message received

"ID1"	: controller byte 1, from integer "000" to "255"
"."	: address separator
"ID2"	: controller byte 2, from integer "000" to "255"
"-"	: channel separator
"CHANNEL"	: channel, an individual channel from integer "01" to "14" channel "00" is not supported by all motor brands. use channel "15" for a group

"-"	: command separator
"ACTION"	: action command available to ALL motor codes, as follows
"up"	: blinds move UP
"dn"	: blinds move DOWN
"sp"	: blinds STOP moving
"gp"	: send blinds to the favourite position
"mu"	: micro-step up and for next-position-up for A-OK motors
"md"	: micro-step down and for next-position-down for A-OK motors
"XX"	: go to a specific position from 01 to 99. ONLY used with motors whose motor code is in this list **"no, db, ra, rb, ap, bl, mb, jo".**

Options only available for motor code “no”:

“i1” : send blinds to intermediate position 1, replaces command “gp”.
“i2” : send blinds to intermediate position 2

Options are only available when using motor code “tb”, this motor code is always for use with the TDBU (TopDown/BottomUp) blinds or for a room with both “tb” and “bf” motor codes.

Rail selector ref. 1: middle rail, 2: lower rail, 3: both rails.
"op” : open shades, move both rails UP
"cl” : close shades, move midle rail UP and lower rail down
"u2” : move lower rail UP
"u3” : move both rails UP
“d2” : move lower rail DOWN
“d3” : move both rails DOWN
“o2” : micro-step lower rail UP
“c2” : micro-step lower rail DOWN
“gp” : send blinds to the favourite position

Options are only available when using motor codes “mb”, “ra”, “ap”, “db”, AND the motors are of the type TDBU (TopDown/BottomUp), otherwise, use the standard set of actions.

Left for the Lower Rail, Right for the Middle Rail.
"op” : open shades, move both rails UP
"cl” : close shades, move midle rail UP and lower rail DOWN
"u2” : move lower rail UP
"u4” : move middle rail UP
"u3” : move both rails UP
“d2” : move lower rail DOWN
"d4” : move middle rail DOWN
“d3” : move both rails DOWN
“x2” : blinds STOP moving (replaces the “sp” for this type of motor)
“gp” : send blinds to the favourite position

“!” : exclamation symbol, command separator preceding the motor code
“MOTORCODE”: code designation to indicate the motor protocol to be transmitted
“bf” : standard Bofu motor
“vb” : vertical Bofu motor
“tb” : top-down/bottom-up Bofu motor
“no” : NEO motor
“rx” : Raex motor

“nc”	: Nice F-Code motors
“nr”	: Nice 0-Code motors
“k1”	: standard A-OK motor
“k2”	: version 2 A-OK motor (Louvolite exclusive)
“k3”	: version 3 A-OK motor (Eclipse exclusive)
“dy”	: standard unidirectional Dooya motor
“by”	: special code compatible with both Bofu and Dooya motors (**depricated!**)
“wt”	: standard Wistar and Alpha motor
“db”	: Bi-directional Dooya motor (has optional TDBU type)
“sy”	: Somfy RTS motor
“gp”	: Gaposa motor
“ra”	: Rollease ARC motor (has optional TDBU type)
“re”	: Rollease EL motor
“ap”	: AMP motors (Turnils) (has optional TDBU type)
“sf”	: Sunfrere motor
“rc”	: Ronco motor
“bl”	: Alta Bliss motor
“mb”	: Coulisse MotionBlinds motor (has optional TDBU type)
“jo”	: Jiecang motor

Examples:

- 123.123-05-up!no	returns: 123.123-05-up
- 012.005-15-dn!bf	returns: 012.005-15-dn
- 129.021-06-87!dy	returns: 129.021-06-87
- abc.def-06-87 (not valid!)	returns: abc.def-06-87
- Test (not valid!)	returns: Test

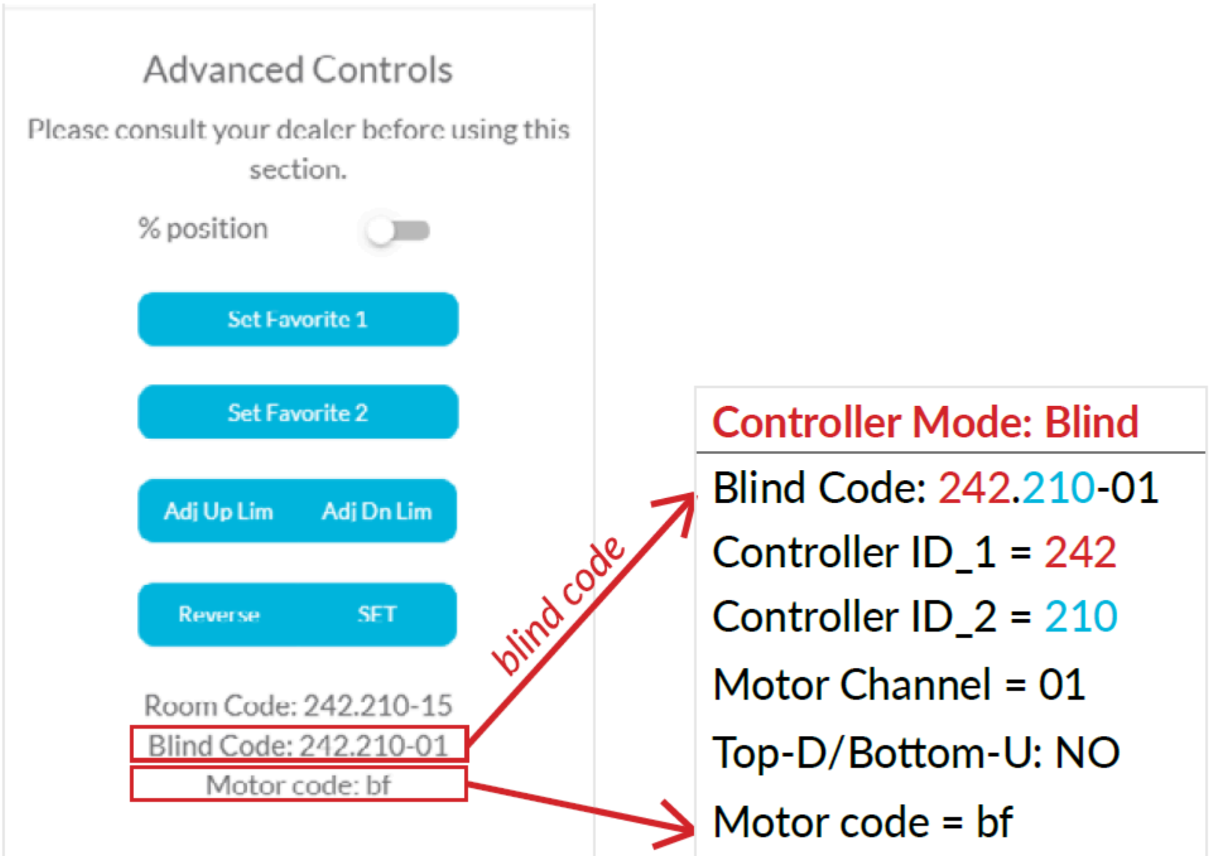
3.1. Finding Controller ID and Channels from APP

Using the app, go to each blind’s advanced page to find the necessary ID numbers. See the image below.

All blinds inside the same ROOM have the same unique combination of Controller ID codes. Use CHANNEL = 15 to control all blinds as a group.

To control individual blinds, use the indicated Motor Channel.

That’s the same behaviour found using physical remote controls.



Date	version	Change log	Approved by
2024/10/28	1.7	- Initial changelog for the document. - Added commands i1 and i2 for motor codes 'no' and 'rx'.	Antonio
2025/07/03	1.8	- Added a new set of actions for “mb”, “ra”, “ap”, “db” exclusively when they are of the type TDBU. - Added indication to the motor codes with the optional TDBU type.	Antonio